

Multiscale Spatial–Spectral Attention Network for Hyperspectral Image Compressed Sensing

Shen Dong¹, Jing Xiao¹, Senior Member, IEEE, Zhen Zhang¹, Huawei Li,
and Liang Liao, Senior Member, IEEE

Abstract—Hyperspectral image (HSI) compressed sensing faces challenges due to spectral redundancy and high dimensionality. Although recent learning-based methods have shown promise, most of them rely heavily on spatial convolutional operations, which are often insufficient in capturing the complex dependencies among spectral bands, thereby limiting the quality of reconstruction. Moreover, due to the use of stripe-like images as input–output pairs, these methods inevitably introduce grid-like artifacts during image stitching. To address these challenges, we propose a novel multiscale spatial–spectral attention network (SSANet) for compressed sensing of HSIs, featuring a newly developed multiscale and multiattention fusion (MMF) module that interactively combines convolutional feature extraction with dual-path attention mechanisms. The MMF module employs a hybrid pooling strategy to effectively model complex spatial–spectral dependencies while maintaining computational efficiency and preserving spectral details. Additionally, the spatial attention (SA) component mitigates grid-like artifacts by enforcing spatial consistency across patch boundaries. Extensive experiments on standard HSI benchmarks demonstrate the superiority of our approach with consistent noise robustness across varying sampling rates. Specifically, at a sampling rate of just 1%, SSANet achieves a 9.5% reduction in spectral angle mapper (SAM) (std = 0.069), a 40.7% reduction in root mean square error (RMSE) (std = 0.245), and a 4% increase in PSNR (std = 0.195) compared to the state-of-the-art methods, validating its effectiveness for HSI reconstruction tasks. The code will be available at: <https://github.com/Shayne2001/SSANet>

Index Terms—Attention mechanism, compressed sensing, deep learning, hyperspectral image (HSI).

I. INTRODUCTION

HYPERSPECTRAL images (HSIs) provide rich spectral information with distinct absorption features across bands, enabling precise land cover classification and water quality estimation [1], [2]. However, their massive data

volume poses significant challenges for storage and transmission [3], [4]. Traditional compression methods, including prediction-based [5], transform-based [6], and vector quantization techniques [7], often struggle to balance compression efficiency with reconstruction quality, especially at low sampling rates.

Hyperspectral compressed sensing (HCS) leverages spectral–spatial redundancies in HSIs to reconstruct data from sub-Nyquist samples, typically using sparse coding or deep learning. Recent advances in learning-based compressed sensing have shown promising results. Recent learning-based methods, such as DCSN [8] and BTCNet [9], enhance robustness via noise-injected training, while attention mechanisms [10] improve feature adaptation and long-range modeling.

DRCT [11] enhances single image super-resolution via dense-residual connections in Swin Transformer blocks, demonstrating that the introduction multiscale spatial information stabilizes windowed attention mechanisms. S^3 RNet [12] proposes a dual-stream network with spatial–spectral attention, demonstrating effective noise suppression through sparse feature weighting. RTCS [13] employs a task-specific training approach, showing exceptional resistance to stripe effects commonly encountered in miniaturized satellites. By utilizing a simplified architecture at the encoder, it reduces the required training samples and enables seamless implementation on integer-8-based encoders. On the decoder side, a two-streamed network with grouped convolution further optimizes efficiency, achieving real-time operation on low-power chips and edge devices.

While recent works like RTCS prioritize extreme lightweight design and task-specific training to suppress stripe artifacts in miniaturized satellite imagery, we pursue an architecture-centric solution that fundamentally models complex spectral–spatial interdependencies within acceptable computational costs. Observing that the existing methods often neglect spectral–spatial coupling during compression, we propose a multiscale spatial–spectral attention network (SSANet) with a multiscale and multiattention fusion (MMF) module. MMF hierarchically captures cross-dimensional interactions through hybrid pooling and multiscale attention, addressing both grid artifacts and spectral distortion. The encoder employs cascaded MMFs with nonsquare convolutions for lightweight compression, while the decoder uses MMF in residual dense blocks for high-fidelity

Received 17 July 2025; revised 16 September 2025; accepted 15 October 2025. Date of publication 22 October 2025; date of current version 3 November 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62202349, Grant 62371351, and Grant 62571397; in part by the Young Elite Scientists Sponsorship Program by China Association for Science and Technology (CAST) under Grant 2023QNRC001; and in part by the Fundamental Research Funds for the Central Universities under Grant ZYTS25036. (Corresponding authors: Jing Xiao; Huawei Li.)

Shen Dong is with the School of Cyber Science and Engineering, Wuhan University, Wuhan 430079, China (e-mail: shaynedong@whu.edu.cn).

Jing Xiao, Zhen Zhang, and Huawei Li are with the School of Artificial Intelligence, Wuhan University, Wuhan 430079, China (e-mail: jing@whu.edu.cn; zhangzhent@whu.edu.cn; lhw@whu.edu.cn).

Liang Liao is with Hangzhou Institute of Technology, Xidian University, Hangzhou 311231, China (e-mail: liaoliang01@xidian.edu.cn).

Digital Object Identifier 10.1109/LGRS.2025.3624612

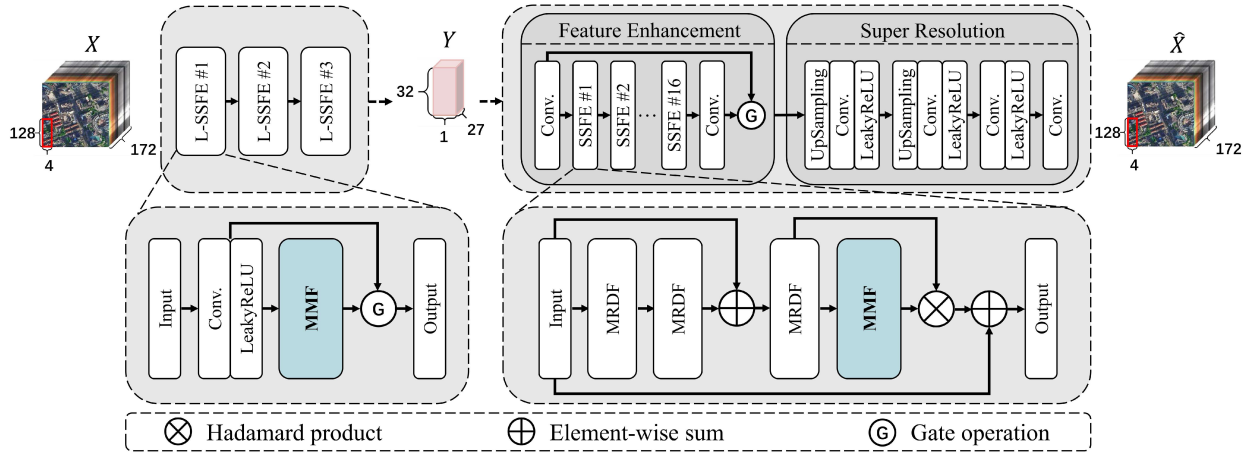


Fig. 1. Network architecture of the proposed SSANet. The encoder compresses hyperspectral data X into the compressed representation Y , and the decoder reconstructs Y into the HSI $\hat{X}$.

reconstruction. Our main contributions are summarized as follows.

- 1) We demonstrate that a lightweight module based on hybrid pooling strategies can effectively model complex spatial-spectral dependencies in HSIs, enabling on-orbit compression and high-quality reconstruction.
- 2) By integrating multiscale spatial attention (SA) with spectral feature fusion, our approach effectively suppresses artifacts induced by stripwise input processing while maintaining computational efficiency.
- 3) We propose SSANet, an encoder-decoder architecture built upon MMF modules, which achieves the state-of-the-art performance in high-fidelity HSI reconstruction even at low sampling rates (e.g., 1%).

II. PROPOSED METHOD

A. Overview of the Proposed SSANet

HSI compressed sensing aims to reduce data redundancy while preserving crucial spectral and spatial information, particularly under the constraints of edge devices with limited computational resources and restricted transmission bandwidth. Formally, let $X \in \mathbb{R}^{H \times W \times C}$ denote an input HSI, where H , W , and C represent the spatial height, spatial width, and spectral bands, respectively. The objective of the compressed sensing task is to establish an efficient mapping

$$Y = \mathcal{E}(X; \Theta_e) \quad (1)$$

where $\mathcal{E}$ is the encoder function, Θ_e denotes the encoder parameters, and $Y \in \mathbb{R}^{h \times w \times c}$ is the compressed representation with $h \ll H$, $w \ll W$, and $c \ll C$. The subsequent reconstruction task is defined as

$$\hat{X} = \mathcal{D}(Y; \Theta_d) \quad (2)$$

where $\mathcal{D}$ is the decoder function and Θ_d denotes the decoder parameters. The objective is to minimize the reconstruction error $\mathcal{L}(X, \hat{X})$ under a target sampling rate.

Due to the strip-by-strip manner for airborne and satellite hyperspectral data acquisition tasks [7] and the limited computational resources at the edge devices, we input strip HSI data

of size $128 \times 4 \times 172$ into a lightweight-designed encoder for feature extraction.

To model the complex spatial-spectral interdependencies efficiently, we propose a dual-stream MMF module. Building upon this, SSANet comprises a lightweight spatial-spectral encoder and a fusion-enhanced decoder, as shown in Fig. 1. The encoder is based on a streamlined version of the spatial-spectral feature extraction (SSFE) module, combining convolutions with the MMF module to meet edge-device constraints. In contrast, the decoder, free from such limits, uses the full SSFE with dense residual connections for high-quality reconstruction from compressed representations.

B. MMF Module

HSIs exhibit nonuniform energy distributions across bands. To adaptively adjust the importance of 172 spectral bands for compressed sensing, we design a hybrid-pooling-based channel attention (CA) architecture combining global average pooling and max pooling [14], preserving both dominant low-frequency information and high-frequency anomalies [6]. However, spatial pooling causes information loss, so we introduce a parallel multiscale SA branch with channel-axis hybrid pooling to complement CA. This branch enforces patch-boundary consistency via learnable gating, mitigating grid-like artifacts from stripwise input fragmentation. The two branches' outputs are adaptively fused through a gating mechanism, compensating for lost spatial details while suppressing artifacts. Together, they form the dual-stream MMF module.

As shown in Fig. 2, the input to the module is denoted as $F_{in} \in \mathbb{R}^{C_1 \times H \times W}$, where C_1 represents the number of channels in the input feature map and H and W represent the height and width of the feature map, respectively. In the initial step of feature processing, the MMF module applies a 1×1 convolutional kernel to reduce the dimensionality of the channels

$$F_{down} = W_{1 \times 1} \cdot F_{in} \quad (3)$$

where $W_{1 \times 1}$ is the weight parameter of the 1×1 convolution kernel and C_2 is the number of channels after dimensionality reduction, where $C_2 = C_1/4$.

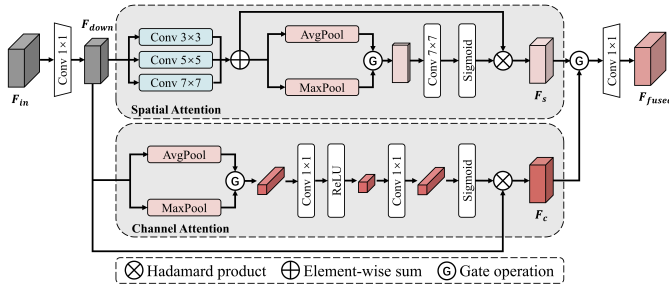


Fig. 2. Network architecture of the proposed MMF block.

For SA, the convolutional kernels of 3×3 , 5×5 , and 7×7 are employed to capture multiscale image information. These features are then fused via an elementwise sum to create a multiscale feature representation

$$F_{\text{multi}} = F_{3 \times 3} + F_{5 \times 5} + F_{7 \times 7} \quad (4)$$

where $F_{k \times k} = \text{Conv}_{k \times k}(X_{\text{down}})$. Next, along the channel axis, we adopt the hybrid pooling strategy to capture both global contextual information and salient feature information, providing a more comprehensive input for generating SA weights.

For CA, both global average and max pooling are applied along the spatial axis, followed by a shared MLP (two 1×1 conv layers with ReLU) to extract channelwise context. This adaptively enhances important channels while suppressing redundant ones through feature response adjustment.

Overall, we obtain F_{fused} in the following manner:

$$F_{\text{fused}} = W_0 (G(\sigma(Q_s K_s^T) V_s), \sigma(Q_c K_c^T) V_c) \quad (5)$$

where W_0 is a 1×1 convolutional upsampling layer, $\sigma(\cdot)$ denotes softmax, and $Q_s, K_s, V_s, Q_c, K_c, V_c$ are the query/key/value tensors for channel/SA, respectively. The gate operation $G(\cdot, \cdot)$ is defined as

$$G(x, y) = \lambda x + (1 - \lambda) y \quad (6)$$

where λ is a learnable parameter.

C. Lightweight Spatial-Spectral Encoder

The encoder employs convolutional layers to progressively reduce input dimensionality, compressing HSI data into low-dimensional features that capture sparse representations (Fig. 1) [14]. To overcome compressive sensing bottlenecks and suppress grid artifacts under limited resources, we propose the lightweight SSFE (L-SSFE) module integrating our MMF module with spatial convolutions. The MMF enhances spatial-spectral feature characterization, indirectly suppressing artifacts while enabling high-fidelity reconstruction through improved compressed representations. Our encoder stacks three L-SSFE blocks, with the j th block processing as follows:

$$\text{L-SSFE}_j(X_i) = G_j(f_j(X_i), \text{MMF}_j(f_j(X_i))) \quad (7)$$

where X_i denotes the i th strip data. The function $f_j(\cdot)$ represents the lightweight spatial feature extraction operation and is defined as

$$f_j(X_i) = \rho_j(X_i \otimes W_j) \quad (8)$$

where the function $\rho(\cdot)$ denotes the convolutional-activation operation, and $W \in \mathbb{R}^{k_s^h \times k_s^w}$ is a nonsquare $k_s^h \times k_s^w$ convolutional kernel designed to align with pushbroom sensor's stripe-like data geometry. The nonsquare kernel (e.g., 3×1) processes narrow input stripes more efficiently than square kernels, avoiding excessive dimensionality reduction while reducing computational overhead and memory usage—crucial for resource-constrained edge devices.

D. Fusion-Enhanced Decoder

In HSIs, certain spectral channels often dominate the representation of hyperspectral pixels, which was generally overlooked by the existing convolutional neural network (CNN)-based SR methods, leading to low reconstruction quality. Therefore, we propose the SSFE module based on MMF, designed to enhance the representational capacity of compressed features by jointly modeling spatial and spectral dependencies.

The decoder adopts square convolutional kernels instead of the encoder's nonsquare design because it processes aggregated features in standard formats rather than stripe-like inputs, and the ground station's relaxed computational constraints enable more complex operations, where square kernels prove more effective for capturing 2-D correlations during spatial-spectral upsampling.

To effectively extract and enhance features, each SSFE block combines three multiscale residual dense feature extraction (MRDF) modules with one MMF module. Inspired by the multiscale feature block (MFB) in DCSN [8], the MRDF employs dense connectivity to allow each convolutional layer to receive inputs from all preceding layers, promoting contextual information reuse and learning of abstract representations. Unlike the original MFB's additive highway connections, MRDF introduces a learnable gating mechanism for adaptive feature fusion. The output is formally defined as

$$F_{\text{dense}} = f_{\text{MRDF}}(X_{\text{in}}, W_{\text{MRDF}}) \quad (9)$$

where f_{MRDF} denotes the function of MRDF, and W_{MRDF} and X_{in} are the sets of the parameters and the input feature map of the MRDF.

As shown in Fig. 1, 16 cascaded SSFE blocks progressively enhance features. The subsequent joint convolutional and upsampling operations then reconstruct spatial and spectral information to generate the final image.

III. EXPERIMENTAL RESULTS

A. Experimental Setup

1) *Dataset*: We use the AVIRIS dataset with 224 spectral bands, reduced to 172 after removing low-quality bands (1–10, 104–116, 152–170, and 215–224) and clipping negative values to zero. A total of 3048 HSI samples of size $256 \times 256 \times 172$ are extracted, with 90% for training and 10% for testing.

2) *Implementation Details*: The model is trained for 20 000 epochs with batch size 36 and initial learning rate $5e^{-4}$, using AdamW optimizer. Spectral angle mapper (SAM)-based

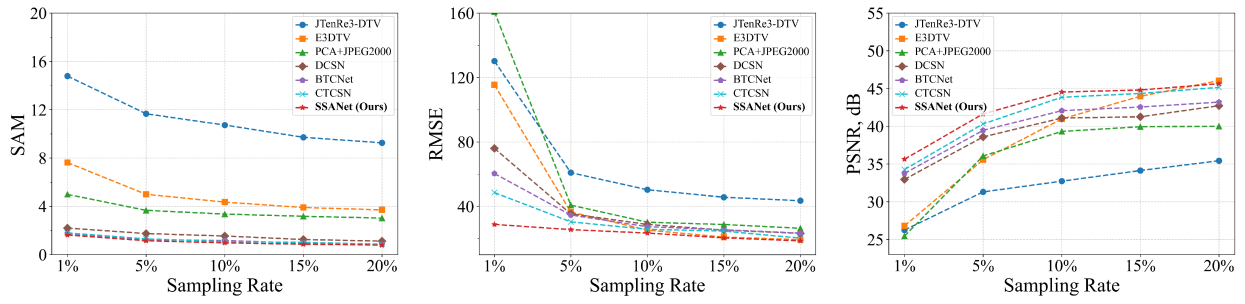


Fig. 3. Performance comparison between the proposed SSANet and other state-of-the-art methods under different sampling rates (from 1% to 20%).

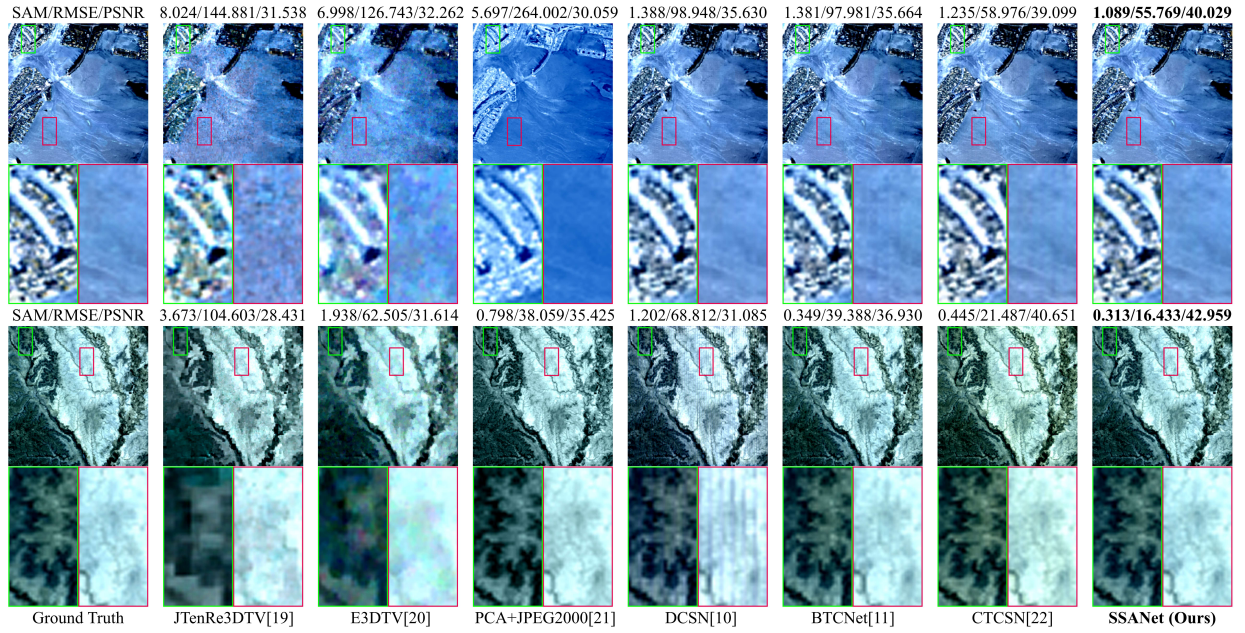


Fig. 4. Visualization results of different methods at 1% sampling rate.

learning rate adjustment is applied every ten epochs. The loss function is defined as

$$\mathcal{L}_1 = \sum_{i=0}^{B-1} \sum_{j=0}^{L-1} \left\| \hat{\mathbf{X}}_j^i - \mathbf{X}_j^i \right\|_1 \quad (10)$$

with B as batch size and $L = W_{\text{in}} \times H_{\text{in}}$ as the number of pixels in each strip of HSI. The reconstruction performance was evaluated using SAM [15], peak signal-to-noise ratio (PSNR) [16], and root mean square error (RMSE) [17]. All experiments are conducted on an NVIDIA A100 GPU.

3) *Baseline Methods*: To illustrate the superiority of SSANet, six SOTA HCS methods are selected as baselines, including JTenRe3DTV [18], E3DTV [19], PCA+JPEG2000 [20], DCSN [8], BTCNet [9], and CTCN [21]. All learning-based baselines were trained from scratch using original hyperparameters on our unified dataset, with enhanced training strategies applied to ensure optimal performance. Each model's results represent the average of five independent training runs under identical experimental conditions.

B. Performance Verification

1) *Quantitative Metric Comparison*: As shown in Fig. 3, our proposed method outperforms other compression sensing methods in terms of all three performance metrics.

TABLE I
COMPUTATIONAL EFFICIENCY AT 1% SAMPLING RATE

Method	Running time (s)	Encoder's Parameters (M)	Encoder's GFLOPs
JTenRe3-DTV	2760.913	—	—
E3DTV	1056.384	—	—
PCA + JPEG2000	2.426	—	—
DCSN	0.867	0.23	0.029
BTCNet (32bit)	2.583	0.23	0.029
CTCSN	1.703	0.18	0.044
SSANet (Ours)	1.473	0.30	0.036

2) *Reconstruction Visualization*: As shown in Fig. 4, we enlarged one high-frequency region (green box) and one low-frequency region (red box) from each method's reconstruction. In high-frequency regions, SSANet best preserves colors and textures. In contrast, the other three deep learning methods exhibit grid-like artifacts in low-frequency regions, while SSANet effectively suppresses them.

3) *Noise Resistance*: Fig. 5 compares the SAM values achieved by the SSANet under different SNR conditions, which demonstrates the noise resistance capability of SSANet.

4) *Computational Efficiency*: As shown in Table I, our SSANet achieves superior performance while maintaining computational overhead within an acceptable range.

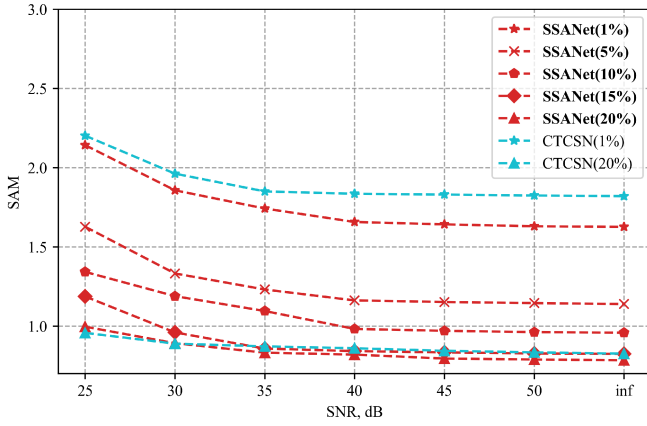


Fig. 5. Performance comparison under different sampling rates and SNRs of the proposed SSANet.

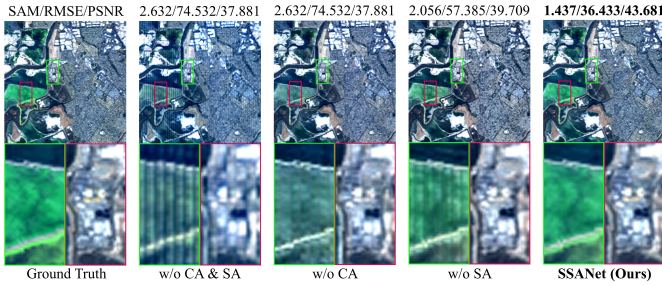


Fig. 6. Visualization results of ablation study at 1% sampling rate.

TABLE II

COMPONENT ANALYSIS OF THE MMF MODULE AT 1% SAMPLING RATE

Models	SA	CA	SAM↓	RMSE↓	PSNR↑
SSANet-N	✗	✗	3.179	75.629	31.267
SSANet-R	✓	✗	2.521	44.173	32.173
SSANet-A	✗	✓	1.753	37.242	33.258
SSANet	✓	✓	1.622	28.790	35.634

C. Component Analysis of the MMF Module

To verify the effectiveness of the proposed MMF module in SSANet, we conducted ablation studies by comparing the full model with three variants: SSANet-N (baseline with ResBlocks, removing both SA and CA), SSANet-R (retaining SA but removing CA), and SSANet-A (using CA but replacing SA with ResBlocks). All models were trained under identical settings. Results in Table II and Fig. 6 demonstrate that the proposed MMF effectively mitigates grid-like artifacts while preserving spectral fidelity.

IV. CONCLUSION

In this letter, we propose an end-to-end network named SSANet, which combines spatial-spectral hybrid attention methods to achieve high redundancy reduction in HSIs and excellent image reconstruction performance at low sampling rates. In the future, we will focus on researching HSI compressed sensing methods with adaptive channel number

capabilities and adaptive sampling rate handling to enhance the model's versatility.

REFERENCES

- [1] H. Grahn and P. Geladi, *Techniques and Applications of Hyperspectral Image Analysis*. Hoboken, NJ, USA: Wiley, 2007.
- [2] Y. Liu et al., "A classification-based, semianalytical approach for estimating water clarity from a hyperspectral sensor onboard the ZY1-02D satellite," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4206714.
- [3] Z. Zhang, J. Xiao, L. Liao, and M. Wang, "RefScale: Multi-temporal assisted image rescaling in repetitive observation scenarios," in *Proc. 32nd ACM Int. Conf. Multimedia*, 2024, pp. 9866–9874. [Online]. Available: <https://openreview.net/forum?id=GVyxNEZLWE>
- [4] H. Wang, L. Liao, J. Xiao, W. Lin, and M. Wang, "Uplink-assist downlink remote-sensing image compression via historical referencing," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5621415.
- [5] G. K. Wallace, "The JPEG still picture compression standard," *Commun. ACM*, vol. 34, no. 4, pp. 30–44, Apr. 1991.
- [6] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, "CBAM: Convolutional block attention module," in *Proc. Eur. Conf. Comput. Vis.*, Sep. 2018, pp. 3–19.
- [7] G. Patané and M. Russo, "The enhanced LBG algorithm," *Neural Netw.*, vol. 14, no. 9, pp. 1219–1237, Nov. 2001.
- [8] C.-C. Hsu, C.-H. Lin, C.-H. Kao, and Y.-C. Lin, "DCSN: Deep compressed sensing network for efficient hyperspectral data transmission of miniaturized satellite," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 9, pp. 7773–7789, Sep. 2021.
- [9] X. Zhou, X. Zou, X. Shen, W. Wei, X. Zhu, and H. Liu, "BTC-Net: Efficient bit-level tensor data compression network for hyperspectral image," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5500717.
- [10] M. Liu, J. Dan, Z. Lu, Y. Yu, Y. Li, and X. Li, "CM-UNet: Hybrid CNN-Mamba UNet for remote sensing image semantic segmentation," 2024, *arXiv:2405.10530*.
- [11] C.-C. Hsu, C.-M. Lee, and Y.-S. Chou, "DRCT: Saving image super-resolution away from information bottleneck," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2024, pp. 6133–6142.
- [12] C.-M. Lee, Y.-F. Lin, L.-W. Kang, and C.-C. Hsu, "Robust hyperspectral image panshaping via sparse spatial-spectral representation," 2025, *arXiv:2501.07953*.
- [13] C.-C. Hsu, C.-Y. Jian, E.-S. Tu, C.-M. Lee, and G.-L. Chen, "Real-time compressed sensing for joint hyperspectral image transmission and restoration for CubeSat," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5512416.
- [14] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7132–7141.
- [15] Y. Zhou et al., "Secchi depth estimation for optically-complex waters based on spectral angle mapping-derived water classification using sentinel-2 data," *Int. J. Remote Sens.*, vol. 42, no. 8, pp. 3123–3145, 2021.
- [16] Q. Huynh-Thu and M. Ghanbari, "Scope of validity of PSNR in image/video quality assessment," *Electron. Lett.*, vol. 44, no. 13, pp. 800–801, Jun. 2008.
- [17] C. Willmott and K. Matsuura, "Advantages of the mean absolute error (MAE) over the root mean square error (RMSE) in assessing average model performance," *Climate Res.*, vol. 30, pp. 79–82, Feb. 2005.
- [18] Y. Wang, L. Lin, Q. Zhao, T. Yue, D. Meng, and Y. Leung, "Compressive sensing of hyperspectral images via joint tensor tucker decomposition and weighted total variation regularization," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 12, pp. 2457–2461, Dec. 2017.
- [19] J. Peng, Q. Xie, Q. Zhao, Y. Wang, L. Yee, and D. Meng, "Enhanced 3DTV regularization and its applications on HSI denoising and compressed sensing," *IEEE Trans. Image Process.*, vol. 29, pp. 7889–7903, 2020.
- [20] Q. Du, N. Ly, and J. E. Fowler, "An operational approach to PCA+JPEG2000 compression of hyperspectral imagery," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2237–2245, Jun. 2014.
- [21] L. Zhang, L. Zhang, C. Song, and P. Zhang, "Hyperspectral image compression sensing network with CNN-transformer mixture architectures," *IEEE Geosci. Remote Sens. Lett.*, vol. 21, pp. 1–5, 2024.